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Studierichting: Informatica
Oefeningenreeks 5D nr. 3.4

$$f(x) = \sec^2 x \text{ en } g(x) = \sec x \cdot \tan x$$

$$\int\limits_{-\frac{\pi}{3}}^{\frac{\pi}{6}} (\sec^2 x - \sec x \cdot \tan x) dx$$

$$= \int\limits_{-\frac{\pi}{3}}^{\frac{\pi}{6}} \left(\frac{1}{\cos^2 x} - \frac{\sin x}{\cos^2 x} \right) dx$$

$$= \int\limits_{-\frac{\pi}{3}}^{\frac{\pi}{6}} \frac{1}{\cos^2 x} dx - \int\limits_{-\frac{\pi}{3}}^{\frac{\pi}{6}} \frac{\sin x}{\cos^2 x} dx$$

$$= [\tan x]_{-\frac{\pi}{3}}^{\frac{\pi}{6}} - [\sec x]_{-\frac{\pi}{3}}^{\frac{\pi}{6}}$$

$$= \frac{\sqrt{3}}{3} + \sqrt{3} - \frac{2\sqrt{3}}{3} + 2 = \frac{2\sqrt{3}}{3} + 2$$

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References